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DNA binding and gene regulatory functions of MSMEG_2295, a repressor encoded by the *dinB2* operon of *Mycobacterium smegmatis*

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Abstract

MSMEG_2295 is a TetR family protein encoded by the first gene of a *Mycobacterium smegmatis* (Msm) operon that expresses the gene for DinB2 (MSMEG_2294), a translesion DNA repair enzyme. We have carried out investigations to understand its function by performing DNA binding studies and gene knockout experiments. We found that the protein binds to a conserved inverted repeat sequence located upstream of the *dinB2* operon and several other genes. Using a knockout of *MSMEG_2295*, we show that MSMEG_2295 controls the expression of at least five genes, the products of which could potentially influence carbohydrate and fatty acid metabolism as well as antibiotic and oxidative stress resistance. We have demonstrated that MSMEG_2295 is a repressor by performing complementation analysis. Knocking out of *MSMEG_2295* had a significant impact on pyruvate metabolism. Pyruvate dehydrogenase activity was virtually undetectable in *ΔMSMEG_2295*, although in the complemented strain, it was high. We also show that knocking out of *MSMEG_2295* causes resistance to H_2O_2, reversed in the complemented strain. We have further found that the mycobacterial growth inhibitor plumbagin, a compound of plant origin, acts as an inducer of MSMEG_2295 regulated genes. We, therefore, establish that MSMEG_2295 functions by exerting its role as a repressor of multiple Msm genes and that by doing so, it plays a vital role in controlling pyruvate metabolism and response to oxidative stress.

INTRODUCTION

Tuberculosis is an age-old disease caused by *Mycobacterium tuberculosis* (Mtb) and related mycobacteria [1]. Although antibiotic-based therapeutic methods are available to control TB, it has not been possible to eradicate it due to antibiotic resistance [2] and tolerance [3]. Researchers have shown that antibiotic resistance can develop in several other organisms due to the accumulation of mutations under stressed conditions [4, 5]. Such mutations occur in bacterial cells due to attempts by DNA polymerases belonging to the Y family to repair damaged DNA in an error-prone manner [6]. Maintenance of genome integrity and viability under the harsh conditions in the macrophages is critical to the intracellular survival of Mtb and, consequently, the pathogenesis that it induces [7]. Hence, understanding how this organism resists DNA damage and survives when threatened by external assaults is crucial for eradicating TB.

DinB (PolIV), along with UmuC and UmuD (PolV), belong to the Y family of DNA polymerases that are capable of executing translesion DNA synthesis and thus repair damaged DNA [6, 8]. The activities of UmuC-D and DinB can increase under circumstances where the SOS pathway is activated either by DNA damaging agents such as UV or beta-lactam antibiotics [9, 10]. Mtb and Msm possess several DinB orthologs, one of which, DinB2, is present in both organisms. Previous studies indicated that DinB2, though it can execute translesion DNA synthesis [11], may not be involved in managing mycobacterial DNA damage [12, 13]. However, the research that we did in our laboratory [14] indicates that DinB2 has a role to play in suppressing thymine-less death, and through this mechanism, it could indeed have a role to play in DNA damage repair. The biological function of mycobacterial DinB2, though, is still unclear. One of the missing links in the story

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Abbreviations: CFU, colony forming unit; EMSA, electrophoretic mobility shift assay; IPTG, Isopropyl ß-D-1-thiogalactopyranoside; Msm, *Mycobacterium smegmatis*; Mtb, *Mycobacterium tuberculosis*; Ni^{2+}-NTA, Ni^{2+}-nitrilotriacetic acid; OD_{600}, optical density; qRT-PCR, quantitative real-time PCR; RNA seq, RNA sequencing.
The RNA sequencing data analysed in this work were deposited and are freely available in the NCBI Sequence Read Archive (SRA) under Bioproject ID PRJNA751760.
Five supplementary tables and five supplementary figures are available with the online version of this article.

is the non-availability of information regarding the induction mechanism of the DinB2 gene expression. If this mechanism is known, we will better understand this enigmatic protein's biological function.

The ORF that precedes *MSMEG_2294* (*dinB2*) is *MSMEG_2295*. In Mtb, counterparts of both MSMEG_2295 and MSMEG_2294 exist, Rv3055 and Rv3056 (DinB2) respectively, and the synteny is maintained. Bioinformatic studies have predicted that Rv3055 belongs to the TetR family of transcriptional regulators. Such studies also predicted that an inverted repeat upstream of *Rv3055* is the site where it binds and executes its function [15]. However, there was no experimental evidence to support the prediction. In Msm the counterpart of Rv3055 is MSMEG_2295. Like Rv3055, it has an N-terminal TetR conserved region indicating that it might also function as a transcriptional regulator. The genes encoding Rv3055 and MSMEG_2295 are present immediately upstream of the corresponding DinB2 coding regions, and, therefore, they might be controlling the expression of the respective *dinB2* genes. However, apart from *dinB2*, these regulators can potentially control the expression of other genes too. It is therefore important to understand how the putative regulators MSMEG_2295 and Rv3055 function.

In this study, we have attempted to characterize the Msm version of Rv3055, which is MSMEG_2295. We show here that MSMEG_2295 controls the expression of multiple genes including *dinB2*, and plays a crucial role in regulating the central carbon metabolism and oxidative stress response in Msm.

METHODS

Bacterial strains and plasmids

The XL1-Blue strain of *Escherichia coli* was used for gene cloning. The T7 RNA polymerase-based expression vector pET28a (Novagen) was used for the over-expression of recombinant genes in *E. coli* BL21 (DE3) [16]. *E. coli* cells were grown in Luria–Bertani (LB) broth with the desired antibiotic. Mycobacterial knockout experiments were performed using the Msm mc^{2}155 derived *ΔDRKIN* strain. In the parent strain mc^{2}155 from which *ΔDRKIN* was derived, a 56 kb region, that includes the cluster *MSMEG_2295/2294/2293*, is duplicated. To avoid the difficulties associated with deleting both copies of *MSMEG_2295,* we used, as proposed earlier, the strain *ΔDRKIN* that is isogenic to mc^{2}155 but lacks one copy of the duplication [17].

For expression in Msm *ΔDRKIN*, we used the acetamide inducible vector pLAM12 [18]. Msm *ΔDRKIN* cells were grown in Middlebrook 7H9 medium (Difco) with 0.01% Tween-80 and 0.2% glycerol. The antibiotics, kanamycin (25 µg ml^{-1}) and hygromycin (50 µg ml^{-1}) were added to the media as and when required. For engineering gene knockouts in mycobacteria, the vectors pGOAL and pNIL were used [19]. The oligonucleotide primers used to amplify genes, and their sequences, are given in detail in Tables S1 and S2 (available in the online version of this article).

Bioinformatic analysis

Protein sequences were retrieved from the NCBI database. The CLUSTALW programme was used for the alignment, and the resulting alignment was used to create phylogenetic trees by the neighbour-joining method with the help of MEGA7 [20]. To predict MSMEG_2295 binding sites in the Msm genome, we used the programme MCAST MEME SUITE [21]. An inverted repeat (IR) containing the sequence 'AACCGGACNNNNGTCCGATT' similar to the one found upstream of the Mtb *dinB2* operon was used as an input motif to search the Msm genome [15]. The output threshold was E-value ≤10.

Purification of recombinant proteins

The target gene (*MSMEG_2295*) was PCR amplified from the genomic DNA of Msm using specific primers and cloned in pET28a using NdeI and BamHI (New England Biolabs, USA) to generate pMP1. A plasmid from which an N-terminal truncated MSMEG_2295 is produced was created and named pMP2. The recombinant vectors pMP1 and pMP2 were transformed into *E. coli* BL21 (DE3). Primer details are given in Table S1. Induction of gene expression was performed using IPTG (0.5 mM) at 37 °C for 3 h. Recombinant His_6-tagged proteins were isolated by Ni^{2+}-NTA agarose (Qiagen, Germany) chromatography under native conditions as per standard protocol [22]. Briefly, the induced cells were harvested and disrupted using a cell disruption equipment (Constant Systems Limited, UK) at 18 kpsi (two pulses) in a lysis buffer [50 mM Tris, 300 mM NaCl, 5 mM β-Me, pH 8]. After centrifugation at 13000 r.p.m. for 30 min, the supernatant was collected, followed by incubation with 1 ml of Ni^{2+}-NTA beads for 30 mins at 4 °C in a gravity column. After collecting the flow-through, it was washed with an increasing concentration of imidazole several times. Finally, elution buffer [50 mM Tris, 300mM NaCl, 5 mM β-Me, 250mM imidazole, 10% glycerol, pH 8] was added, and eluates of 1 ml each were collected. The eluates were analysed by 12% sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). Fractions containing pure protein were pooled and dialysed overnight at 4 °C against 50 mM Tris-HCl, 300 mM NaCl, 5mM β-Me pH 8 containing 15% glycerol. The quality of the purified protein was judged by performing 12% SDS- PAGE gel electrophoresis.

Expression of recombinant genes in mycobacteria

The target genes were amplified from Msm and cloned in the expression vector pLAM12 (primer details given in Table S1) using NdeI and NheI. The recombinant vectors pMP3 and pMP4 expressing the wild-type MSMEG_2295 and its N-terminal truncated version respectively were introduced into electrocompetent mycobacterial cells as described previously [14], and transformants were selected on kanamycin containing MB7H9 agar plate. For induction of gene expression, cells were grown overnight to an OD_{600} of 0.3 and induced with 0.2% acetamide for 3 h.

Electrophoretic mobility shift assay (EMSA)

EMSA was performed as described previously [23]. A 108 bp DNA fragment harbouring the *dinB2* promoter was amplified using the primers DinPro_FP and DinPro_RP (Table S1) and labelled with [γ^{32}P] ATP (purchased from BRIT, Mumbai, India) using T4 polynucleotide kinase (PNK) (Thermo Scientific, USA). The PCR product was purified using the Qiagen PCR purification kit and used as a probe for EMSA reaction. The reaction mixture (30 µl) contained 3 µl of 10× binding buffer [100 mM Tris (pH 8), 600 mM NaCl, 30 mM $MgCl_2$, 1 mM EDTA, 1 mM DTT, 20% glycerol], 1 µg of salmon sperm DNA, 1 nM of labelled probe (10000 cpm.) and the recombinant MSMEG_2295 protein. The reaction mixtures were preincubated for 15 min and then incubated for an additional 15 min on ice after the probes were added. The DNA-protein complexes were separated on a 5% native PAGE gel by electrophoresis in the 0.5 Tris-borate buffer [50 mM Tris-borate, 1 mM EDTA] at 130V for 3 to 4 h at 4 °C after a prerun at 100V for 1 h. Following electrophoresis, the gel was dried on Whatman paper, and the bands were visualized by autoradiography. Competition binding experiments were performed by including molar excess of unlabelled competitor DNA fragments during the preincubation period.

DNase I footprinting assay

DNase I footprinting analysis was performed as described previously [24]. Binding reactions with the [γ^{32}P] labelled DNA probe (1 nM) were carried out under the conditions mentioned in the case of EMSA. After the reaction, 50 ng DNase I (Sigma) was added, and the mixture was incubated at 37 °C for 3 min. The digestion was stopped by adding 2 µl of 0.5 mM EDTA. The digested fragments were then purified using the phenol-chloroform extraction method and then alcohol-precipitated. Finally, the samples were washed with 70% ethanol, dried and re-suspended in loading buffer [98% (v/v) deionized formamide, 10 mM EDTA, 0.025% (w/v) xylene cyanol and 0.025% (w/v) bromophenol blue] followed by boiling for 5 min and rapid chilling. The DNase I digested fragments were then separated by running a sequencing gel at 1300 V for 3.5 to 4 h. The gel was dried and exposed to Kodak Bio-Max film. An A+G ladder was prepared using 1 nM of labelled DNA by limited exposure of the DNA to formic acid followed by piperidine treatment.

Gene knockout experiments

A two-step recombinational protocol was used for generating unmarked disruption of target mycobacterial genes [19]. For the first step, a suicide vector was constructed using the knocked-out version of the gene. The knockout was done by amplifying the flanking regions of the segment marked for deletion. Primer details are given in Table S2. The PCR products were digested with NheI, the site for which was introduced into the internal primers. The NheI digested fragments were ligated. The ligated DNA was then used as the template for PCR using flanking primers. The PCR copy obtained was cloned in the p2NIL vector. Marker cassette from pGOAL19 was further incorporated to construct the suicide vector. Both the knockouts were performed in the Msm *ΔDRKIN* strain, which, unlike Msm, has one copy of the *dinB2* operon instead of two [17]. Msm *ΔDRKIN* electrocompetent cells were electroporated with the respective suicide constructs. The Single Cross Over (SCOs) mutants were selected on agar plates containing kanamycin, hygromycin, and X-gal. The SCOs (kan^R, hyg^R, blue) were grown in MB7H9 agar to allow the second crossover event. Serial dilutions of the culture were then plated on solid media containing X-gal and 2% sucrose to enable phenotypic counterselection of putative DCO (Double Cross Over) recombinant mutants. The suc^R and white colonies were selected for further analysis. Knockout was confirmed by performing PCR analysis followed by DNA sequencing using a 3130X genetic analyser (Applied Biosystems).

RNA isolation and quantitative real-time PCR

Total RNA was isolated from Msm cells using commercially available kits (Qiagen RNeasy kit) using the protocols mentioned by the manufacturer. Gene-specific reverse primers were used for first-strand cDNA preparation from 1 µg of total RNA using Revert Aid Reverse Transcriptase (Thermo Scientific). For 25 µl reaction, 50 ng cDNA, 0.125 mM dNTPs, 0.1 pmol µl^{-1} of each primer, and 0.25 U of Q5 NEB High-Fidelity DNA polymerase were used for end-point RT-PCR. Quantitative Real Time PCR (qRT-PCR) was performed using Power SYBR Green PCR Master Mix (Applied Biosystems) using the Applied Biosystems 7500 Fast Real-Time PCR System. The C_t values obtained were normalized against those of the 16S rRNA gene as described earlier [25]. The DNA sequences of the primers used for RT-PCR and qRT-PCR are given in Table S1. In all the RT related experiments we always performed a RT (-) control in which RT (Reverse Transcriptase) was omitted. This was done to ensure that the samples were free from DNA contamination.

RNA-seq analysis

The extracted RNA samples were shipped to the Neuberg Centre for Genomic Medicine, Ahmedabad, India for the whole transcriptome analysis. The company performed the RNA-seq analysis, and data is provided back on a secure server hosted by the company. In brief, the submitted RNA was depleted using Ribocop rRNA depletion kit using the manufacturer's protocol (Lexogen #125–127, Greenland, NH, USA). In total 100 ng of ribodepleted RNA was utilized and converted into the final library using Truseq stranded RNA library preparation kit following manufacturer's protocol (Illumina, USA). Minimum eight million high-quality paired-end reads were generated from each sample. Raw reads were trimmed for adapter removal, and high-quality reads were utilized for the differential gene expression calculations using the DESeq2 tool. Final statistical analysis was performed, and TPM (transcript per million) values were determined. The statistical significance was expressed as *P* values. The reference genome used for TPM calculation was that of *Mycobacterium smegmatis* mc^{2}155 (NC_008596.1). The results obtained from two biological repeat experiments were analysed by creating a

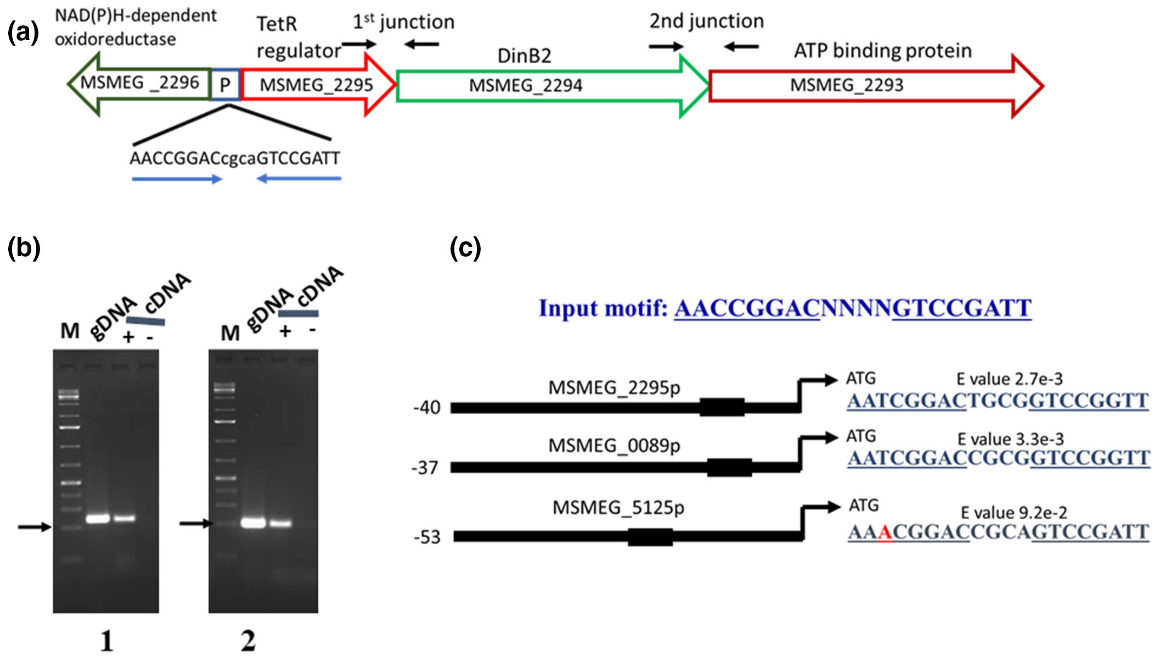

Fig. 1. The *dinB2* operon. (a), The structure of the *dinB2* operon is shown along with the location of an inverted repeat sequence in the promoter region (P) between the divergently oriented genes *MSMEG_2295* and *MSMEG_2296* that serve as the binding site of MSMEG_2295. (b) Agarose gel electrophoresis of PCR products obtained using the primers (marked by arrows) that span the first (1) and second (2) junctions between the three genes as indicated in (a). Black arrows in (b) corresponds to 200 bp (in the ladder). The amplicons were derived either from the genomic DNA (gDNA) or from the polycistronic RNA (cDNA). (c) The binding sites for MSMEG_2295 predicted using MCAST MEME SUITE having top three scores in the genome of Msm are shown along with the input sequence. A list of all the matches obtained is given in the Fig. S1. Mismatch to the consensus is highlighted in red.

volcano plot (Log_{10} *P* value change against Log_2 fold change) using GraphPad prism software.

Biochemical assays for metabolites and enzymes

Pyruvate concentration in cell-free extracts was estimated by reducing it to lactate using the enzyme lactate dehydrogenase and measuring the extent to which NADH the co-substrate was oxidized in the process. The reaction was carried out in 50 mM Tris-HCl (pH 7.5), 50 mM $MgCl_2$, 0.4 mM NADH (SRL) and 4.5U lactate dehydrogenase (SRL). Oxidation of the NADH to NAD^+ was monitored spectrophotometrically at 340 nm for 3 mins. The rate of change of OD_{340} per minute (ΔOD_{340} min^{-1}) was considered the measure of pyruvate concentration. Background correction was done by deducting the value obtained from a control assay performed without the enzyme. Determination of the pyruvate dehydrogenase (PDH) complex activity in cell-free extracts was achieved by measuring the amount of NADH that a fixed amount of the extract could convert to NAD (ΔOD_{340} min^{-1}) in the presence of pyruvate and coenzyme A (CoA) [26]. The reaction was carried out by the addition of 2 mM pyruvate (SRL) and 0.13 mM CoA (Sigma) to an assay mixture containing, 1 mM $MgCl_2$, 0.2 mM thiamine diphosphate (ThDP) (Sigma), 2.5 mM NAD^+ (SRL), 2.6mM cysteine-HCL (Sigma), 50 mM potassium phosphate (pH 7) and 20 µl cell-free extracts in a total volume of 100 µl. Background corrections were done by deducting the value obtained from a control assay without the addition of pyruvate.

Determination of pyruvate dehydrogenase E1 (PDH E1) subunit activity in cell-free extracts was achieved by an assay that measures the rate of reduction of the artificial electron acceptor 2,6-dichlorophenolindophenol (DCPIP) by the E1 component, with pyruvate as a substrate (DCPIP assay) [26]. The reaction was initiated by adding pyruvate (final concentration of 400 µM) in a suspension comprising 0.2 mM ThDP, 2 mM $MgCl_2$, 50 µM DCPIP (Merck), 100 mM potassium phosphate (pH 7), and cell free extract followed by monitoring the decrease of DCPIP absorbance at 600 nm. Background corrections were done by deducting the value obtained from a control assay without the addition of pyruvate.

RESULTS

The *dinB2* operon

To obtain insight into the regulation of the DinB2 gene (*MSMEG_2294*) expression, we analysed the DNA sequence of the *dinB2* locus. The analysis results revealed that three genes, *MSMEG_2295*, *MSMEG_2294* (*dinB2*), and *MSMEG_2293*, are clustered together and arranged in the same order as mentioned indicating that these are likely to form an operon

(Fig. 1a). To confirm the operonic structure of this locus, we synthesized primers bridging the two junctions between the three genes and performed PCR both with the genomic DNA and RNA. In both the cases (Fig. 1b), we obtained amplicons of the desired size, and hence it appears that these three genes constitute a polycistron. Upstream of this operon, there exists an inverted repeat sequence, very similar to the one found at an equivalent location in the case of Mtb, that could serve as a promoter-operator for the operon (Fig. 1a) [15]. The promoter is likely to be a divergent one controlling the expression of another gene, *MSMEG_2296* expressed in the opposite direction. The precise functions of the proteins expressed by the genes other than *MSMEG_2294* (*dinB2*) are unknown. MSMEG_2295 has been predicted to be a TetR family protein [15], MSMEG_2293 a possible ATP binding protein, and MSMEG_2296 a NADPH-dependent oxidoreductase.

DNA sequences similar to the inverted repeat located upstream of the *dinB2* operon are also present at two other sites in the Msm genome, one of which is located upstream of *MSMEG_0089* and the other *MSMEG_5125* (Fig. 1c). Apart from these two, we found several other less significant sites in the genome of Msm, that could potentially act as binding sites for MSMEG_2295 (Fig. S1).

Binding of MSMEG_2295 to the *dinB2* operon promoter-operator

The organization of the *dinB2* operon (Fig. 1a) strongly suggests that MSMEG_2295 is likely the regulator of this operon. Moreover, in an earlier *in silico* analysis of TetR family transcription factors (TFTR) of *Mycobacterium tuberculosis* (Mtb), it was predicted that Rv3055, the Mtb counterpart of MSMEG_2295, has a binding site upstream of its gene [15]. Based on this information, we proceeded to perform a DNA binding assay using a purified version of the recombinant protein, MSMEG_2295 (Fig. S2a). The upstream region of the *MSMEG_2295* gene was labelled with [γ^{32}P] ATP and used for DNA binding assays. The results of the EMSA assays performed (Fig. 2a) revealed that the protein indeed binds to the promoter region in a dose-dependent manner. The binding was specific as it could be competed out efficiently by an unlabelled self-competitor DNA and a related DNA sequence present upstream of the gene encoding the protein MSMEG_0089. However, an unrelated piece of DNA of the same length did not compete (Fig. 2b). Not only *MSMEG_0089*, but a DNA fragment representing the upstream region of *MSMEG_5125* also competed efficiently (Fig. 2c), indicating that these are all legitimate MSMEG_2295 binding sites. We also created an N-terminal deleted version of the protein in which the putative DNA interactive region (amino acid residues 50–69) was removed (Fig. S2a, b). When we used the mutated version in EMSA assays, we did not get any binding (Fig. 2d). This observation further indicates that the DNA-protein interactions are specific. We also performed DNaseI foot-printing experiments using the radiolabelled probe and identified precisely the binding site (Fig. 2e). The binding site spans the inverted repeat situated upstream of *MSMEG_2295* (Fig. 2f). Interestingly we found that the site was present downstream of the reported transcriptional start sites (TSS, Fig. 2f), although conventionally it should be upstream [27]. Thus, it appears that MSMEG_2295 could be functioning by a road blocking mechanism rather than promoter occlusion.

MSMEG_2295 is a repressor

The results presented above indicate that MSMEG_2295 could function as a repressor. To investigate this, we created an unmarked knockout mutant of this gene (Fig. S3). We then assessed the expression levels from the *dinB2* operon and *MSMEG_0089* in the knockout strain (*ΔMSMEG_2295*) and compared it to the parent, Msm (*ΔDRKIN*). To examine the effect of the *MSMEG_2295* knockout on the expression from the *dinB2* polycistron, we used the same primers that we used to prove that the gene cluster *MSMEG_2295/2294/2293* forms an operon. From the results obtained (Fig. 3), we conclude that transcription from the first junction increased significantly (Fig. 3c), whereas from the second did not (Fig. 3d). The results indicate a possible polar effect of the knockout of *MSMEG_2295*, resulting from premature transcription termination. Alternatively, internal controls elements may exist that controls *MSMEG_2293* expression. However, the increase in the *MSMEG_0089* expression was much higher (about 100-fold compared to 4.5 for the *dinB2* operon first junction) (Fig. 3e).

Complementation experiments to demonstrate repression by MSMEG_2295

The results presented in the previous section indicate that MSMEG_2295 might be functioning as a repressor. To confirm this issue further, we complemented *ΔMSMEG_2295* by expressing the wild-type copy of the gene from an inducible vector (*ΔMSMEG_2295/pMP3*) and monitored the consequences by performing RNA analysis. Firstly, we analysed whether the complementing gene was being overexpressed or not. The results (Fig. 4a, b) show that in the case of the knockout strain (*ΔMSMEG_2295*), we could not detect any *MSMEG_2295* transcripts, which we expected. In the (*ΔMSMEG_2295/pMP3*) cells, we observed that expression could be seen with and without induction of gene expression. Thus, *MSMEG_2295* expressed itself in both the induced and the uninduced states, although it was higher in the induced as expected (Fig. 4b). We then ascertained the impact of *MSMEG_2295* expression on that of *MSMEG_0089* in the *ΔMSMEG_2295* strain. The results indicated that expression of *MSMEG_2295* even at the basal level inhibited transcription of *MSMEG_0089* by over eighty-sevenfold, and following induction, one hundredfold (Fig. 4c, d, lanes marked *ΔMSMEG_2295/pMP3* U/I as compared to *ΔMSMEG_2295*). At the same time, when we used the N-terminal deletion mutant, we did not see any repression (Fig. 4c, d, lanes marked *ΔMSMEG_2295/pMP4* U/I as compared to *ΔMSMEG_2295/pMP3*) even though, in this case too, we see that the corresponding gene expressed itself adequately (Fig. 4a, b).

RNA-seq analysis indicates that multiple genes are regulated by MSMEG_2295

We have also analysed the global effect of knocking out *MSMEG_2295* by performing an RNA-seq analysis. The results

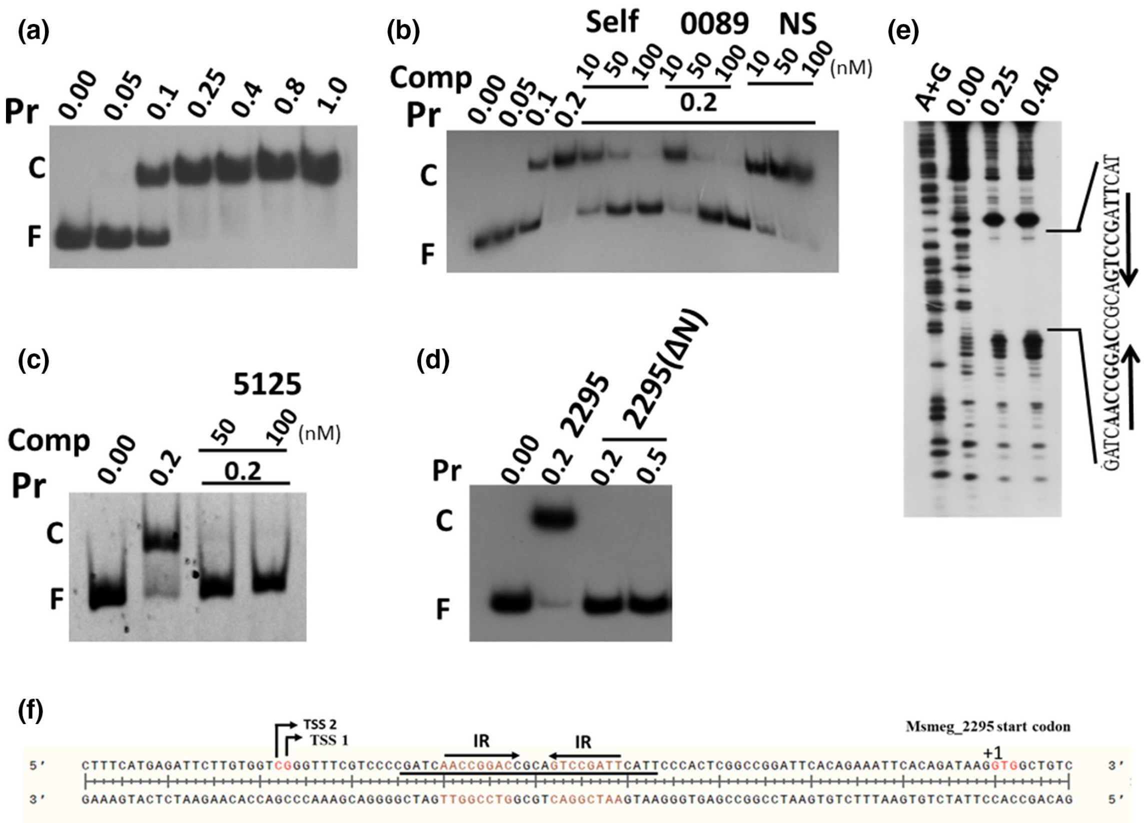

Fig. 2. Operator binding activity of MSMEG_2295. The promoter-operator region of the *dinB2* operon (f) was used as a probe for EMSA (a–d) or DNaseI foot-printing (e). The concentrations (µM) of the protein (Pr) used, either the wild-type (2295) or the N-terminal deleted version (MSMEG_2295(ΔN)) are shown above. In (b, c), the molar concentrations of the unlabelled competitor (Comp) used are indicated. The concentration of 50 nM is equivalent to 50-fold molar excess. The competitors used were the *MSMEG_2295* binding sites located upstream of the *dinB2* operon (self), *MSMEG_0089* (b) and *MSMEG_5125* in (c). In (b) a 100 bp fragment derived from the mycobacteriophage D29 genome was used as a non-specific competitor (NS). (d) The binding activity of MSMEG_2295(ΔN) mutant version was compared with that of the wild-type. In (e) as in (a – d), the protein concentrations are mentioned on the top. The lane corresponding to the A+G ladder is marked. The sequence over which the footprint is observed is shown on the right. The sequence of the DNA segment used for the footprinting assay is shown below (f). The location of the inverted repeat sequence to which the protein binds is indicated with inverted arrows. The MSMEG_2295 footprint is underlined. The transcription start sites of the *dinB2* operon as mapped in an earlier investigation [27] are indicated (TSS) along with the initiating codon of the first ORF (MSMEG_2295).The nucleotide residues that constitiute the TSSs, the IR and the start codon are highlighted. The first nucleotide of the start codon is indicated as (+1).

presented in the form of a volcano plot (Fig. 5, Table S3) reveal that five genes were induced with a high probability, the *P* values being lesser than 10^{-50}. Out of these five, three showed an increase by more than tenfold. The maximum gain (82-fold) was observed in the case of *MSMEG_0089*, followed by *MSMEG_5125* (58-fold) and *MSMEG_2296* (16-fold) (Table S3). In the upstream region of all these genes, the consensus binding site for MSMEG_2295 is present. We did expect an increase in transcript level corresponding to *MSMEG_0089* based on the results of the RT-PCR experiments described above. The increase in the level of *MSMEG_5125* was an additional finding. The expression of this gene should increase as it has an MSMEG_2295 binding site in its promoter region. The new result we obtained was the high-level transcription of the divergently expressed gene *MSMEG_2296* (16-fold) even though in the opposite direction increase was only 1.8-fold for *dinB2* and none for *MSMEG_2293* (Table S3). The conclusions that we draw are (a) MSMEG_2295 controls expression from the *dinB2* operon promoter-operator region in a bidirectional manner and (b) the results obtained with the *dinB2* operon do not genuinely reflect the maximum extent to which its induction can occur. It should have been to the extent comparable to that of *MSMEG_2296*. As mentioned above, manipulation within *2295*, the first gene in the operon, must have negatively affected the expression of the downstream genes.

We obtained a piece of important information from the RNA-seq experiment: the expression of the genes *MSMEG_5124* and *MSMEG_0090* located downstream of *MSMEG_5125* and *MSMEG_ 0089*, respectively, but in the complementary strand also increased significantly (Fig. S4, Table S3). Neither of these genes possesses the canonical MSMEG_2295 binding sites in their promoter regions. Hence it is an enigma how these genes became a part of the MSMEG_2295 regulon.

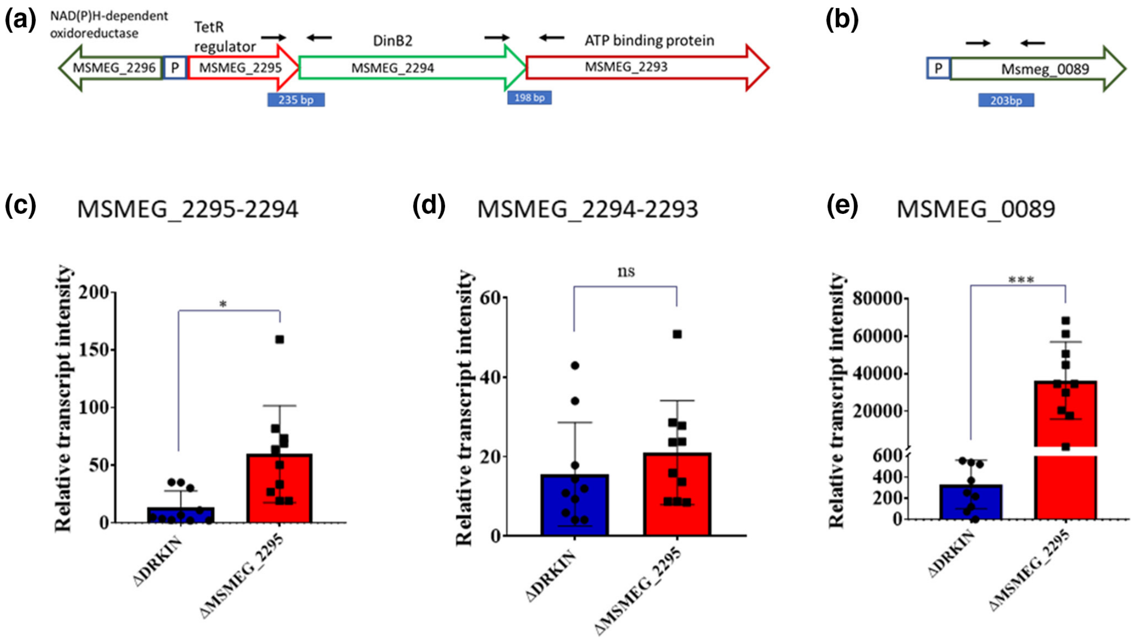

Fig. 3. Differential gene expression in the *ΔMSMEG_2295* strain (red) relative to the parent *ΔDRKIN* (blue) as monitored by performing qRT-PCR. Three loci were chosen, two of which (c, d) correspond to the junction regions between the three genes of the *dinB2* operon as indicated (a). The other target used was *MSMEG_0089* (e), the promoter region of which (b) harbours a sequence that is nearly identical to the MSMEG_2295 binding site located upstream of the *dinB2* operon. The converging arrows in (a) and (b) denote the primers used for PCR. The comparison was done between the transcript levels of the targets relative to that of the 16S rRNA gene (relative transcript intensity). The experiment was performed ten times. All the data points are shown along with bars representing the mean values±standard deviation. The *P* values are indicated by astericks (* for *P* value<0.05, and ***<0.001). Non significant differences are indicated as ns.

MSMEG_2295 controls pyruvate catabolism

To understand the biological implications of MSMEG_2295 mediated repression, we sought to investigate the function of MSMEG_0089, the expression of which increased by almost one hundredfold in the *ΔMSMEG_2295*. The databases annotate MSMEG_0089 as a pirin family protein. A literature survey revealed that pirin family proteins might have diverse functions, including a role in regulating pyruvate dehydrogenase activity [26] as is the case with the pirin$_{Sm}$, a *Serratia marcescens* pirin protein. Sequence alignment and phylogenetic analysis reveal that MSMEG_0089 is homologous to pirin$_{Sm}$ (28.48%) [26] and another from *Streptomyces ambofaciens,* PirA, (66%) [28] that is also believed to be able to influence the central carbon metabolism. We were thus interested in investigating whether MSMEG_0089 and its regulator MSMEG_2295 have any influence on pyruvate metabolism of Msm.

Pyruvate is converted to acetyl CoA by the enzyme pyruvate dehydrogenase complex. Several subunits, E1, E2, and E3, comprise the PDH complex, of which E1 catalyses the first step in which the enzyme complex decarboxylates pyruvate [26, 29]. The PDH inhibitor pirin$_{Sm}$, to which MSMEG_0089 appears to be related (Fig. S5), has been demonstrated to inhibit PDH activity by blocking the functioning of the E1 subunit [26]. We anticipated that a block in the pyruvate dehydrogenase step of the central pathway for carbon metabolism would increase the pyruvate level in the cells. To examine this possibility, we first created the knockout mutant *ΔMSMEG_0089*. We then compared the pyruvate content in this strain to the parent, *ΔDRKIN*, and also *ΔMSMEG_2295*. The fundamental difference between *ΔMSMEG_2295* and *ΔMSMEG_0089* is that whereas in the latter, we expect the level of MSMEG_0089 to be nil; in the former, it should be maximum as the expression of its gene is derepressed. In addition to these strains, we used a complemented strain in which we introduced a plasmid vector pMP3 expressing *MSMEG_2295* into *ΔMSMEG_2295* (*ΔMSMEG_2295/pMP3*). The cells were grown to the mid-log phase in all the cases, followed by harvesting and analysis.

The results reveal that the average pyruvate content of the *ΔMSMEG_0089* strain was lower than that of the parental strain Msm *ΔDRKIN* (Fig. 6a, compare blue striped to green hatched box). Except for one data point, which we have analysed and found to be an outlier, the values corresponding to the others were less than in *ΔDRKIN*. This result indicates that MSMEG_0089 has a role in promoting the pyruvate level by reducing its conversion to acetyl CoA. However, what was interesting was the degree of difference observed in the pyruvate levels of the *ΔMSMEG_2295* and *ΔMSMEG_0089* strains. In *ΔMSMEG_2295*, in which the expression of *MSMEG_0089* we expected to be maximal, the pyruvate content was significantly higher than in *ΔMSMEG_0089,* in which we know for sure that MSMEG_0089 is absent (Fig. 6a compare purple checker board box to green hatched). When we performed

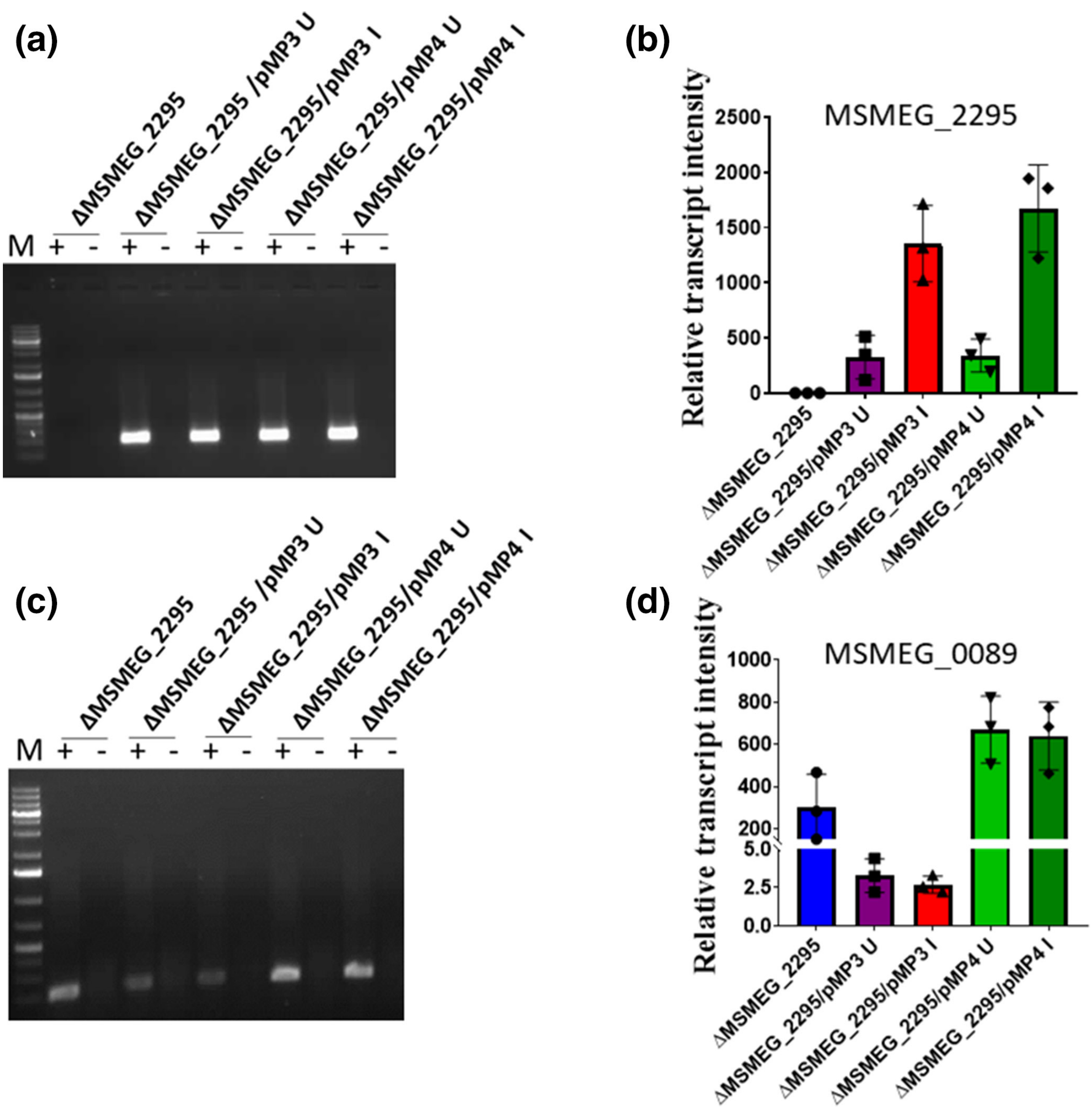

Fig. 4. Repressor activity of MSMEG_2295. The knockout strain (*ΔMSMEG_2295*) was transformed with vectors that overexpress either the wild-type version of MSMEG_2295 (pMP3) or the N terminal sequence deleted version of the protein (pMP4). The transcripts produced from MSMEG_2295 coding sequences in the absence (U, uninduced) or presence (I, induced) of the inducer, acetamide were monitored by performing end point PCR (a) and qRT-PCR (b). The same RNA samples were also analysed for the presence *MSMEG_0089* specific RNA by end-point (c) as well as qRT-PCR (d). The experiment was performed thrice. Transcript intensities (mean of three biological repeat experiments±standard deviation) were derived relative to those of 16S rRNA (relative transcript intensities) and plotted for the individual strains as indicated. In these experiments, we performed a RT (-) control to ensure that in the RT(+) lanes the product is derived from the RNA and not contaminating DNA.

the complementation experiment; we found that in the *MSMEG_2295* complemented strain (*ΔMSMEG_2295/pMP3*); the pyruvate levels were significantly reduced (Fig. 6b compare purple checker board box to red boxes). The results indicated that MSMEG_2295 plays a role in pyruvate metabolism apparently by controlling the transcription of *MSMEG_0089*.

To confirm this observation, we performed a similar experiment as above by assaying the activity of the pyruvate dehydrogenase complex (PDH) and its E1 subunit (Fig. 6c, d). The results show that in the *ΔDRKIN* strain, the PDH and PDH E1 activities were low (Fig. 6c, d, blue striped boxes). In the *ΔMSMEG_2295* strain, the activities were either as low (PDH, Fig. 6c, blue compared to purple) or lower than *ΔDRKIN* (PDH E1, Fig. 6d). In fact, in the case of PDH E1, the activity in *ΔMSMEG_2295* strain was undetectable in all the experiments. In the *ΔMSMEG_0089* strain, the activities were higher relative to *ΔDRKIN* and *ΔMSMEG_2295* (Fig. 6c, d, green hatch box compared to blue striped and purple

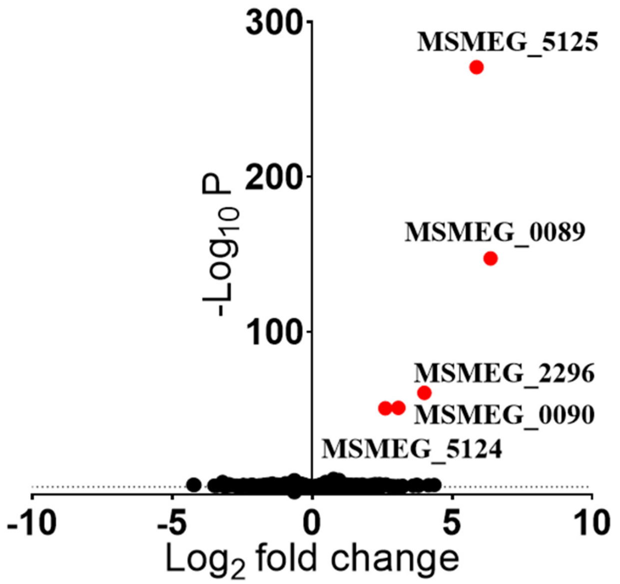

Fig. 5. Volcano plot representation of RNA-seq analysis data. The RNA-seq data obtained from the RNA sequencing of *ΔDRKIN* and *ΔMSMEG_2295* was compared by plotting the -$Log_{10}P$ values against Log_2 fold change. The expression of genes that was significantly higher in *ΔMSMEG_2295* as compared to *ΔDRKIN* are indicated by red symbols. The actual values are mentioned in Table S3.

checker box), which is consistent with the results obtained in the pyruvate level assays.

We investigated what happens to the PDH and PDH E1 activities of the *ΔMSMEG_2295* strain upon complementation. The results show that in the complemented strain (*ΔMSMEG_2295/pMP3*), the level of PDH activity was significantly higher than in *ΔMSMEG_2295* (Fig. 6c, d, red grid box compared to purple). The activity observed in *ΔMSMEG_2295/pMP3* was not only comparable to that of *ΔMSMEG_0089*, but in the induced state (I), the average PDH and PDH E1 activities were higher (Fig. 6c, d, red grid box compared to green). Although this difference was not statistically significant, we think that there is a possibility that not just MSMEG_0089, but products of other genes belonging to the MSMEG_2295 regulon may contribute to the phenotype observed.

MSMEG_2295 knockout cells are resistant to superoxide

The results presented above indicate that one of the consequences of the deletion of *MSMEG_2295* is that a significant decrease in the PDH activity occurs. A decrease in PDH activity is likely to decrease the efficiency with which the TCA cycle operates, decreasing oxidative metabolism and stress. Previous studies using antibiotics have indicated a link between increased TCA cycle activity and oxidative stress [30]. Moreover, it has been shown in *Streptomyces* [28] that knocking out of the gene for a pirin protein that we found is an ortholog of MSMEG_0089 (Fig. S5) resulted in increased sensitivity to oxidative stress. We, therefore, predicted that MSMEG_2295 might have a role to play in the management of oxidative stress. To investigate this aspect, we compared the sensitivity of *ΔMSMEG_2295* and the parental *ΔDRKIN* strain towards oxidative stress induced by H_2O_2. We found that whereas *ΔDRKIN* was sensitive to H_2O_2 (20 mM), *ΔMSMEG_2295* was resistant (Fig. 7a, b, blue striped boxes compared to purple, and Table S4). In the next experiment, we evaluated the oxidative stress sensitivity of *ΔMSMEG_2295* relative to the complemented strain (*ΔMSMEG_2295/pMP3*). In this case, we find that whereas the *ΔMSMEG_2295* strain was resistant to 10 mM of H_2O_2, the complemented strain with or without induction of gene expression was sensitive (Fig. 7c purple checker board box compared to red diagonal up and grid box). Thus, although there was a day-to-day variation in absolute sensitivity, the results indicate that *ΔMSMEG_2295* is relatively more resistant than the strains which express the wild-type version of the gene. The differential effects were evident from the growth monitoring experiment (OD_{600}) (Fig. 7a, c) as well as measurement of viability (c.f.u. ml^{-1}) (Fig. 7b, d, Table S4). We found that the impact on growth was much more when we measured the c.f.u. ml^{-1} in comparison to OD_{600}.Thus the viability counts (c.f.u. ml^{-1}) dropped by more than 2000-fold in *ΔDRKIN* but no change in *ΔMSMEG*_2295 (Fig. 7b). In the complementation experiment, we found that induced expression of MSMEG_2295 resulted in 863-fold decrease in viability, whereas in its absence the change was negligible (less than five-fold). (Fig. 7d, Table S4). Thus, we establish that MSMEG_2295 has a role to play in oxidative stress management.

Plumbagin acts as an inducer of the MSMEG_2295 regulon

The MSMEG_2295 protein belongs to the TetR family of regulators. Antibiotics and various organic molecules control the activity of many of these regulators [31, 32]. We thus expected that antibiotics might modulate the action of MSMEG_2295. To test this possibility, we selected the antibiotics novobiocin, INH, and rifampicin. INH and rifampicin are well-known anti-TB drugs. We chose novobiocin as in a previous study, induction of the *dinB2* operon by this antibiotic was reported in Mtb [33]. In addition, we also used plumbagin which has antimycobacterial properties [22]. The results show that while novobiocin and plumbagin abolished the DNA binding activity of MSMEG_2295 the others did not (Fig. 8a). We then proceeded to examine whether these compounds induce the expression of *dinB2*. In the case of novobiocin, we did not get significant induction above the basal level. But, upon using plumbagin (5µg ml^{-1}, a sublethal dose), we observed an average threefold increase in the expression of *dinB2* and fourfold of *MSMEG_0089* (Fig. 8b, c).

DISCUSSION

The gene for DinB2 (MSMEG_2294) is located within a cluster comprising the ORFs *MSMEG_2295/2294/2293*. The protein databases predict that the first ORF in this cluster (MSMEG_2295) is a TetR family protein. We first wanted to know whether these genes constitute an operon. To answer this question, we performed PCR with primers spanning the

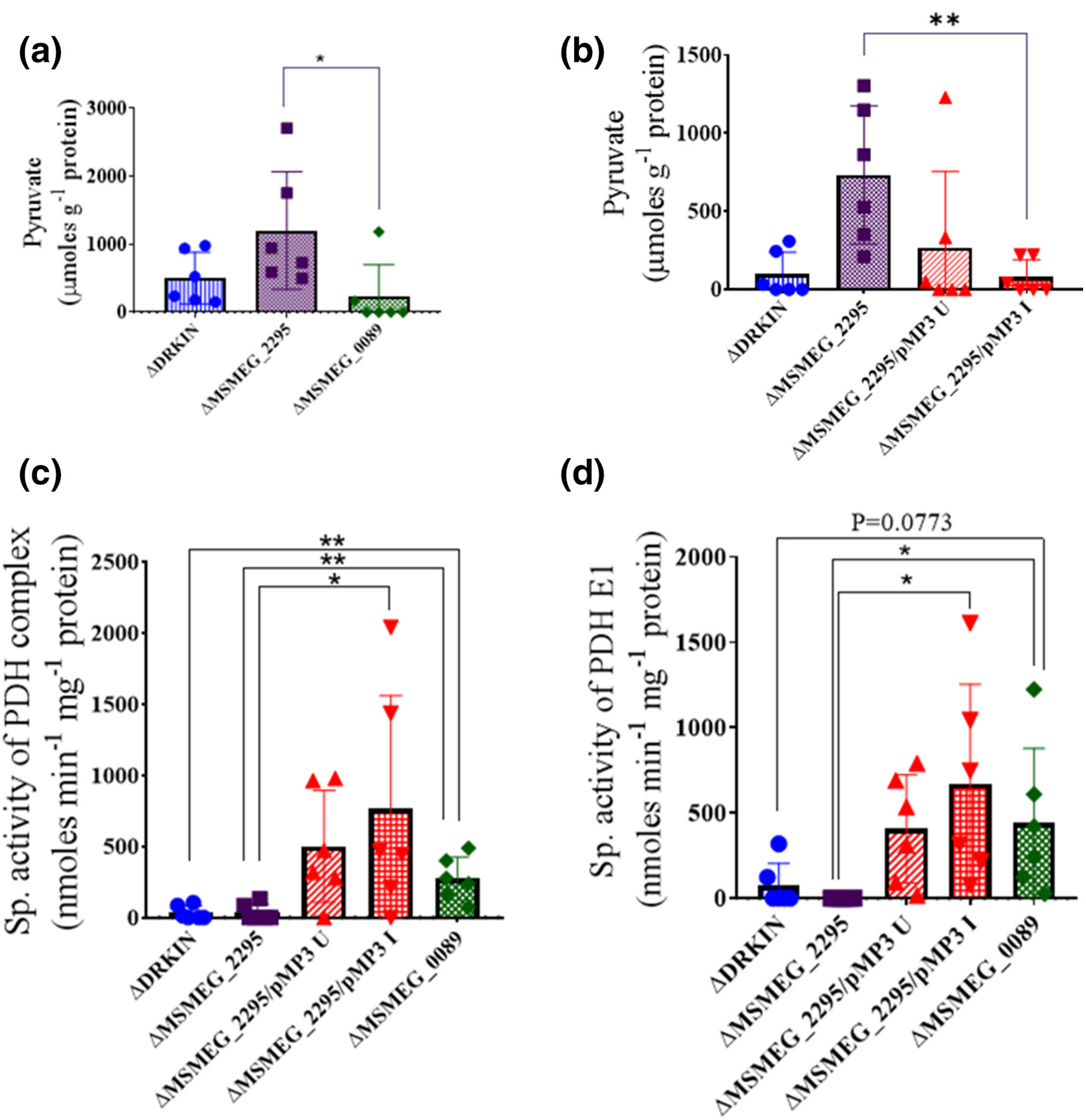

Fig. 6. Effect of *ΔMSMEG_0089* and *ΔMSMEG_2295* knockout on pyruvate metabolism of Msm. The pyruvate contents of the strains *ΔMSMEG_2295* and *ΔMSMEG_0089* strains were compared with each other and the parental strain Msm (*ΔDRKIN*) (a). The effect of complementation of *ΔMSMEG_2295* (*ΔMSMEG_2295/pMP3*) on pyruvate content was also examined (b). In (c, d) pyruvate dehydrogenase complex and pyruvate dehydrogenase E1 activities in the same strains was assessed. In the complementation experiments (b–d) U and I stand for uninduced and induced expression respectively. The experiments were performed six times and the data are presented as the mean of six results±SD. The significant differences are indicated, * for $P<0.05$ and ** $P<0.01$. In one instance where the *P* value was marginally higher than 0.05, the exact value is mentioned.

junctions. Since the PCR results were positive, we conclude that this cluster constitutes a polycistron.

We then wanted to test whether MSMEG_2295 controls the expression of the operon. To achieve our goal, we first performed DNA-protein interaction studies using the promoter region of the *dinB2* operon. We show that an inverted repeat element similar to the one predicted to be the binding site of Rv3055 (the Mtb counterpart of MSMEG_2295) was the target [15]. We could find similar binding sites upstream of two other genes, *MSMEG_0089* and *MSMEG_5125*. Next, we created an *MSMEG_2295* knockout (*ΔMSMEG_2295*) and tested the extent to which induction of the *dinB2* operon takes place using PCR-based methods. For this purpose, we used the junction primers expected to yield amplification products derived from the polycistronic RNA. The results show that in *ΔMSMEG_2295*, only the transcript level spanning the first junction, between *MSMEG_2295* and *MSMEG_2294*, increased significantly by about 4.5-fold but not the second. RNA-seq analysis supported the results obtained using the PCR method. In *ΔMSMEG_2295*, the expression of *MSMEG_2294*

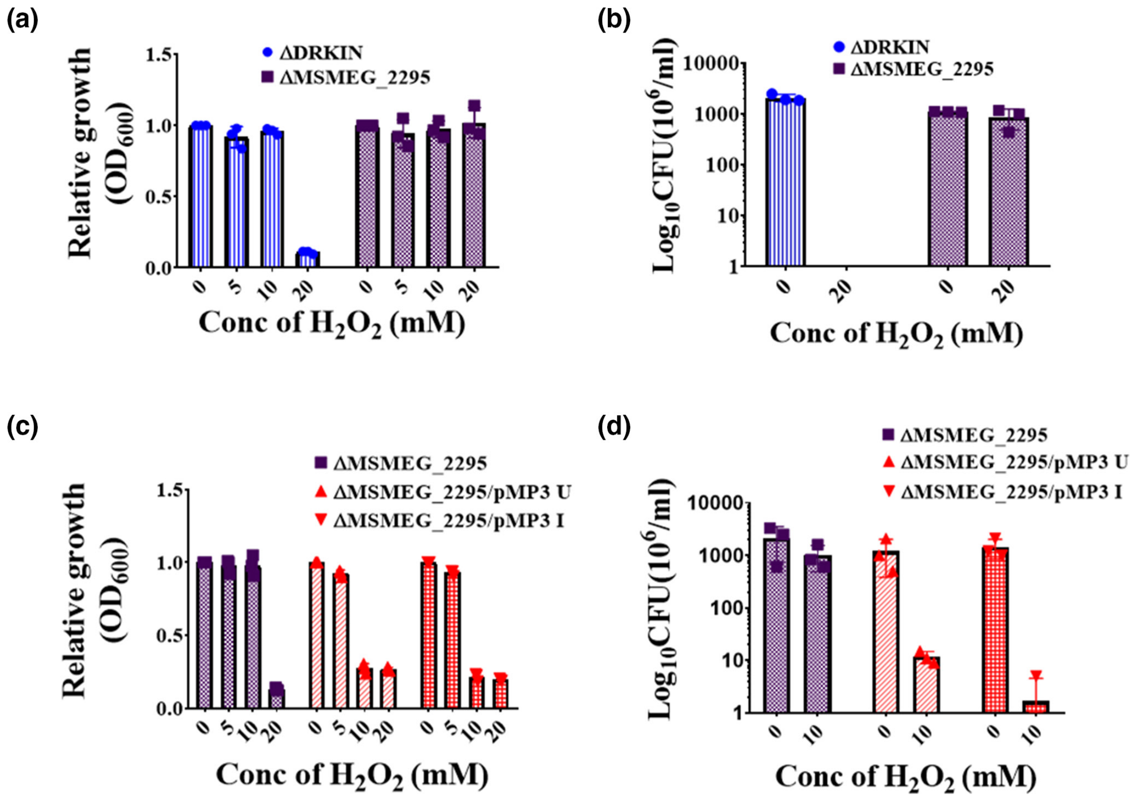

Fig. 7. Effect of H_2O_2 treatment on the growth of *ΔDRKIN*, the knockout mutant *ΔMSMEG_2295* that was derived from it and the complemented strain *ΔMSMEG_2295/pMP3* with (I) or without (U) induction of gene expression. The growth was monitored by measuring OD_{600} (a, c). The values were expressed relative to the OD_{600} of the sample that was untreated (0 mM). The cultures in which significant differences in relative growth was observed between the strains tested were further analysed by performing viability assays (c.f.u. ml^{-1}) (b, d). The exact c.f.u. values derived from the cultures are mentioned in Table S4.

(*dinB2*) increased by about twofold but that of *MSMEG_2293* did not change.

Interestingly, the RNA-seq data revealed that *MSMEG_2296*, the gene expressed divergently from the same regulatory region as the *dinB2* operon, was induced in the *MSMEG_2295* knockout to a much larger extent, sixteen-fold. The relatively low induction in the expression level of *dinB2* compared to *MSMEG_2296* is possibly due to a polar effect that may have resulted from the manipulations done in the first gene of the operon *MSMEG_2295*. The observation that an increase in transcript intensity from the second junction region was almost nil in the *MSMEG_2295* knockout, although that from the first was higher, three to fourfold, supports the possibility of a polar effect.

Apart from *MSMEG_2294*, we also examined another gene, *MSMEG_0089*, upstream of which we could locate a consensus MSMEG_2295 binding sequence. Quantitative RT-PCR analysis showed that the expression of *MSMEG_0089* increased remarkably, by almost one hundredfold. We could efficiently reverse the high-level expression of *MSMEG_0089* in *ΔMSMEG_2295* by expressing a wild-type copy of *MSMEG_2295* but not a mutant version lacking the coding region for the N-terminal TetR conserved domain. These results prove that MSMEG_2295 represses gene expression by binding to its cognate sequence.

The RNA-seq analysis supported the qRT-PCR data, but in addition, it indicated that several other genes, apart from *MSMEG_0089*, were highly expressed in *ΔMSMEG_2295*. The list includes *MSMEG_2296*, mentioned above, and also *MSMEG_5125*. The analysis also revealed that the genes located downstream of *MSMEG_5125* and *MSMEG_0089*, *MSMEG_5124*, and *MSMEG_0090*, respectively, were co-induced. It is an enigma how this happened as these genes are located on the complementary strand and therefore not part of the same transcriptional unit as their respective opposite strand partners. Moreover, they do not seem to possess canonical MSMEG_2295 binding sites in their promoter regions. One possibility could be that these genes may be having non-canonical MSMEG_2295 binding sites in their promoter regions, the identity of which we are unaware. Alternatively, there could be a looping mechanism by which the operator-bound MSMEG_2295 can exert long-range repression on neighbouring genes.

The functions of most of the genes that belong to this regulon have been predicted but not experimentally proved. The pair

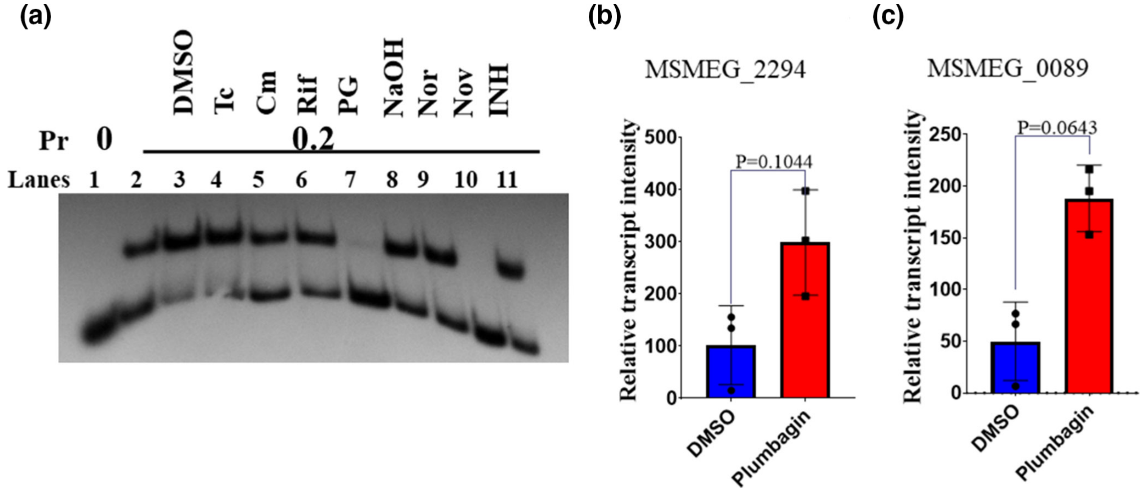

Fig. 8. Effect of antibiotics on the DNA binding activity of MSMEG_2295 (a) and expression of genes *MSMEG_0089* and *MSMEG_2294* controlled by it (b, c). Binding assays were performed using EMSA in the presence or absence of the equimolar concentrations (1 mM) of the antibiotics tetracycline (Tc), chloramphenicol (Cm), rifampicin (Rif), plumbagin (PG), novobiocin (Nov), norfloxacin (Nor) and isoniazid (INH).The protein concentration used was 0.2µM (lanes 2 to 11). Mock treatments were performed with either DMSO or NaOH as indicated. Quantitation of *MSMEG_2294* and *MSMEG_0089* specific transcripts was done by performing qRT-PCR analysis of RNA extracted from plumbagin (5 µg ml^{-1}) or vehicle (DMSO) treated early log phase (OD_{600} of 0.5) *ΔDRKIN* cells, after 3 h of treatment (b, c). Transcript intensities (mean±standard deviation of three biological repeat experiments) were derived relative to that of the 16S rRNA (relative transcript intensity). The *P* values are mentioned in b) and c) as they are marginally higher than .05.

MSMEG_5125–5124 is likely to be involved in fatty acid metabolism as MSMEG_5125 is a probable isoprenoid binding protein and MSMEG_5124 a 2, 4 dienoyl reductase. Another pair is *MSMEG_ 0089–0090*. MSMEG_0090 belongs to a family of transport proteins that gives resistance to fusaric acid, whereas MSMEG_0089 may be controlling central carbon metabolism as it belongs to the pirin family, several members of which have this capacity [26, 28]. The central carbon metabolism has a role to play in antibiotic sensitivity and the associated induction of oxidative stress [34]. Therefore, we speculate that the combination of MSMEG_0089 and MSMEG_0090 may have a role to play in antibiotic resistance.

A previous study revealed that bacterial pirin family proteins could control the central carbon metabolism by inhibiting the activity of pyruvate dehydrogenase, as has been shown in the case of the Serratia protein pirin$_{Sm}$ [26]. We suspected that MSMEG_0089 being a pirin family protein it may have a similar function. Knock-out of the gene encoding MSMEG_0089 resulted in significant pyruvate metabolism changes - pyruvate level decreased and PDH activity increased. Interestingly in the *ΔMSMEG_2295* strain, the opposite was observed, pyruvate level increased, and PDH activity decreased. We also observed that in the MSMEG_2295 complemented *ΔMSMEG_2295* strain, the pyruvate level returns to the basal level, and at the same time, the PDH as well PDH E1 subunit activity increases significantly. The results indicate that MSMEG_2295 controls pyruvate metabolism by regulating the expression of *MSMEG_0089*. However, when we attempted to complement the *ΔMSMEG_0089* by overexpressing *MSMEG_0089* in it, no significant inhibitory effect on the PDH enzyme activity was observed. Thus, it could be that for its PDH inhibitory function, MSMEG_0089 requires assistance from other members of the MSMEG_2295 regulon. Alternatively, it may be that MSMEG_0089 itself is not the inhibitor of PDH activity but some other functionally related protein, the production of which is adversely affected in *ΔMSMEG_0089*. We can also not rule out issues like lack of adequate expression or synthesis of the active form of the protein in the complemented strain. However, in the case of MSMEG_2295, there is no ambiguity. The results presented convincingly demonstrates that MSMEG_2295 has a role to play in controlling pyruvate metabolism, apparently by regulating genes, the products of which inhibit PDH activity.

There is an intricate link between pyruvate metabolism, and the operation of the TCA cycle as pyruvate is one of the sources of acetyl CoA. An increase in TCA activity results in increased oxidative metabolism, which can, in turn, cause oxidative stress as it happens in the case of antibiotic treatment [30]. Moreover, an earlier study has shown that a pirin family protein from *Streptomyces*, which we think is an ortholog of MSMEG_0089, controls the central carbon metabolism and sensitivity to oxidative stress [28]. Since MSMEG_2295 exerts control on pyruvate metabolism, therefore, we investigated the sensitivity of the *ΔMSMEG_2295* strain to the oxidative stress induced by H_2O_2 and compared with ΔDRKIN. We find that *ΔMSMEG_2295* is resistant to the bactericidal action of H_2O_2, possibly because, in these cells, oxidative metabolism is low. We could reverse the resistance phenotype by complementing *ΔMSMEG_2295*, indicating that *MSMEG_2295* and no other gene caused the resistance phenotype. Thus MSMEG_2295 has a significant role to play in oxidative stress resistance mechanisms.

We have also attempted to find an inducer for MSMEG_2295. Out of a series of anti-mycobacterials tested, we found that plumbagin and novobiocin effectively abolished the DNA binding activity of MSMEG_2295. Out of these two, we could demonstrate that plumbagin can induce the MSMEG_2295 controlled genes *dinB2* and *MSMEG_0089* by about three to fourfold. Interestingly, there is no significant difference in the extent to which induction of *dinB2* and *MSMEG_0089* took place, even though the difference was considerable in the *MSMEG_2295* knockout experiment. Thus, the degree to which MSMEG_2295 regulates expression from *MSMEG_0089* and *dinB2* may not be much different, certainly not to the extent indicated in the *MSMEG_2295* knockout experiment. We were unable to demonstrate inducibility with novobiocin, although in Mtb, it happens [33]. It may be that the PCR-based assay system used was not efficient in detecting small changes in the RNA level. Alternatively, it could be that there is some mechanism in Msm that prevents novobiocin from acting as an inducer.

The most important achievement in this study is that we have demonstrated that MSMEG_2295 functions as a repressor of several genes, the products of which are likely to have essential metabolic roles, management of oxidative stress being one of them. However, we cannot extrapolate the results we obtained using Msm to Mtb at this stage as we found several differences. Thus, we could not trace the existence of the orthologs of MSMEG_5125, MSMEG_0089, and MSMEG_0090, the genes for which express themselves highly in the *MSMEG_2295* knockout mutant (Table S5). Also, the number of Mtb genes that possess the consensus binding site for Rv3055, the counterpart of MSMEG_2295, is limited. Only the *dinB2* operon has a significant site, but none that are outside it. Msm, therefore, appears to use the regulator MSMEG_2295 more extensively as compared to Mtb.

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Conflicts of interest

The authors declare that there are no conflicts of interest.

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